

Problem 1: Put a penny on each gray square of strip A. The pennies are the pegs. A jump: a peg hops over a peg right next to it, in a straight line, into the empty square just beyond, and the peg it hopped over comes off the strip. Make one jump. Draw an X in the square where the peg lands.

Strip A

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Problem 2: On strips B, C and D, put pennies on the gray squares and try to make a jump. Circle *possible* or *not possible* for each strip. On strip D, draw an X in every square a peg could land on with one jump.

Strip B

circle one: possible not possible

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Strip C

circle one: possible not possible

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Strip D

circle one: possible not possible

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Problem 3: On each of strips E to J, put pennies on the gray squares. Jump until no jump is possible, in either direction. Write the number of pegs left. Then set the strip up again and jump in a different order. Write the fewest pegs you managed to leave.

Strip E

pegs left, first try: _____ fewest: _____

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Strip F

pegs left, first try: _____ fewest: _____

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Strip G

pegs left, first try: _____ fewest: _____

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Strip H

pegs left, first try: _____ fewest: _____

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Strip I

pegs left, first try: _____ fewest: _____

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Strip J

pegs left, first try: _____ fewest: _____

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Problem 4: Strip K is empty. Draw dots in five of its squares to make a pattern of five pegs that can be cleared down to one peg. Check it with pennies. On strip L, draw a pattern of five pegs that always gets stuck with three or more pegs left, and check that too.

Strip K

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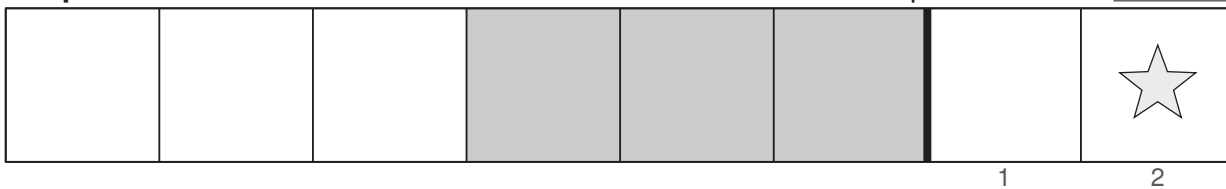
Strip L

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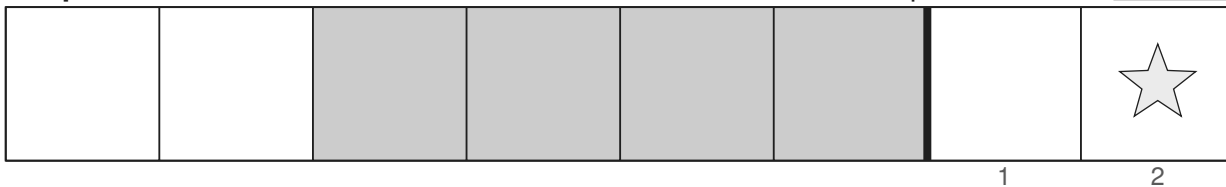
Problem 5: On strip M the thick line is the river, and the squares to the left of it are the home side. Put pennies on the two gray squares. Cross the river with one jump. Draw an X where the peg lands. Strips N, O and P start with three, four and six pegs on the home side. For each strip, put pennies on the gray squares, jump as many times as you like, and try to get a peg onto square 2, the square with the star. Write the farthest square any peg reached, 1 or 2.

Strip M**Strip N**

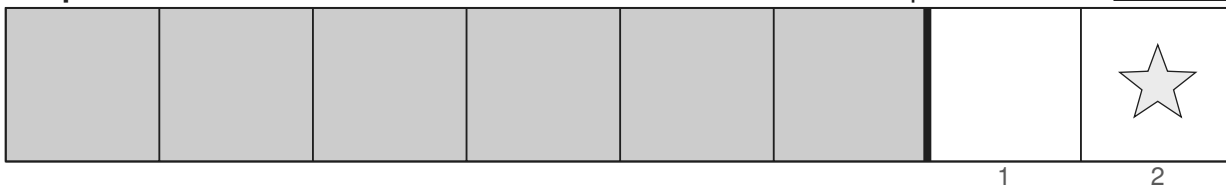
farthest square reached: _____

**Strip O**

farthest square reached: _____

**Strip P**

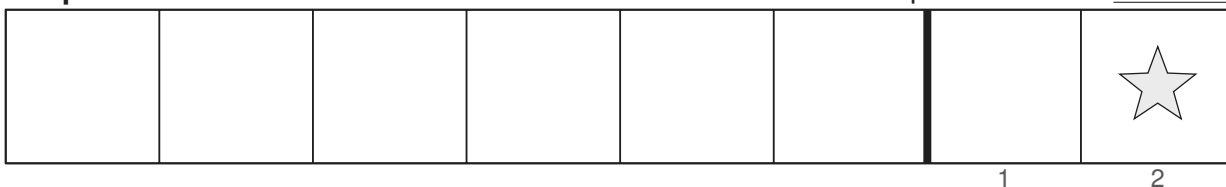
farthest square reached: _____



Problem 6: On strip Q, choose your own starting squares on the home side, up to six pennies. Draw a dot in each square you used. Then play and write the farthest square reached.

Strip Q

farthest square reached: _____



Problem 7: Strip S has a value printed in the corner of each square. The total of a position is the sum of the values of the squares that have pegs on them. For each row of the table: put pennies on the squares listed, write the total, make the jump described, and write the new total. In the last column write *up*, *same* or *down*.

Strip S

1	1	2	3	5	8	13	21
							
						1	2

pegs on the squares valued . . .	total before	total after	up, same or down
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all six home squares; make the only jump that is possible			
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2 and 3; jump to the left			
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2 and 3; jump to the right			
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5 and 8; jump to the right, across the river			
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8 and 13, one on each side of the river; jump to the left			
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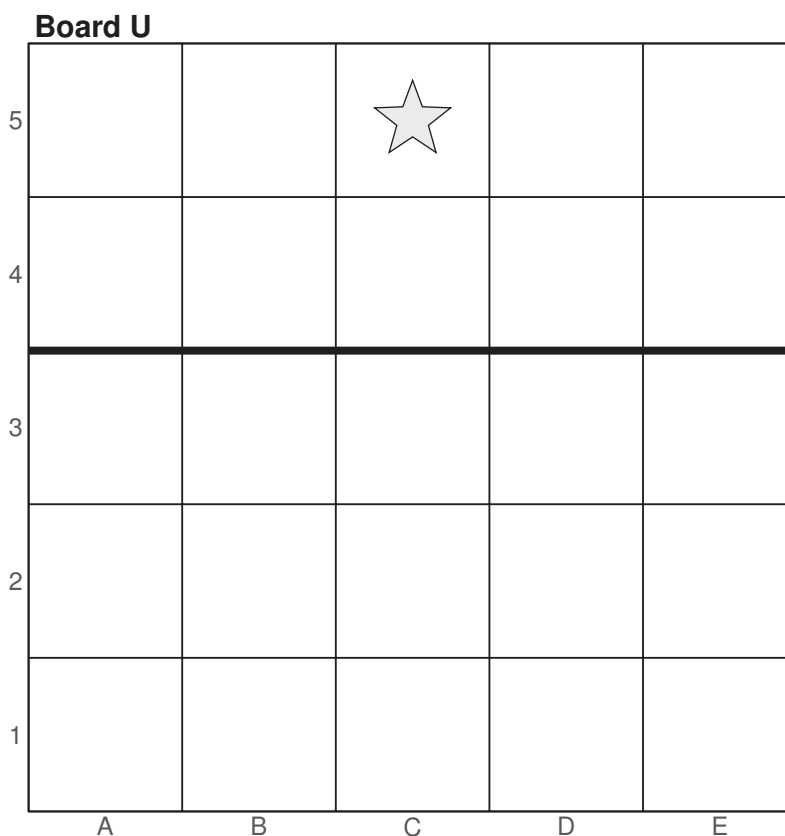
Problem 8: Write the total of the six home squares of strip S: _____ and the value of square 2: _____. Look at the last column of your table. Tell the adult at your table why a peg can never land on square 2 of strip S, whatever jumps you make.

Problem 9: Strip T has seven home squares and no values. Write values in its corners so that the pattern of strip S continues from the left: each value is the sum of the two values before it. Write the total of the seven home squares: _____ and the value of square 2: _____. Say why no peg can reach square 2 of strip T either.

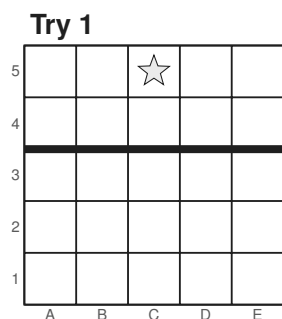
Strip T

							
						1	2

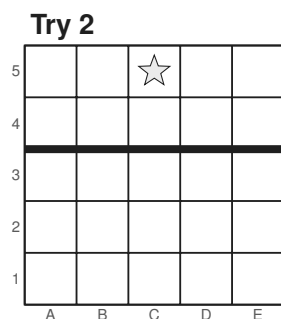
Problem 10: Board U is a grid with the river across it. Squares are named by column letter and row number, like C3. On a board, jumps go up, down, left or right, never diagonally. Put pennies on C3 and C2. Cross the river with one jump. Write the name of the square where the peg lands: _____



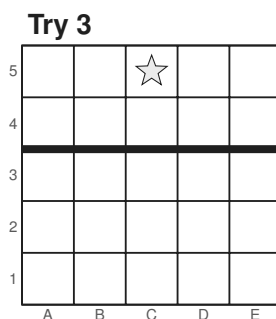
Problem 11: Use board U again. Put only three pennies on home squares of your choice and try to get a peg to the star at C5. Try at least four different placements. Record each one on a small board below: draw a dot in each square you used and write the highest row number any peg reached.



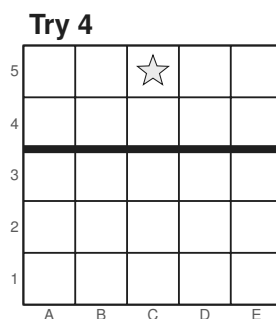
highest row: _____



highest row: _____



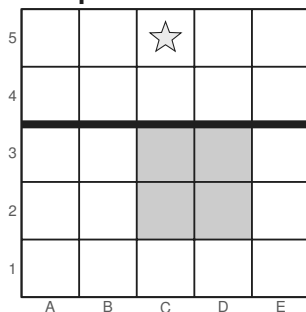
highest row: _____



highest row: _____

Problem 12: Each small picture shows a shape of four pegs on the home side of board U. For each shape, copy it onto board U with pennies and try to get a peg to the star at C5, jumping as many times as you like. Under the picture write *yes* if a peg reached the star and *no* if it never did. For each shape that works, also write its jumps in order, like C1 over C2 to C3.

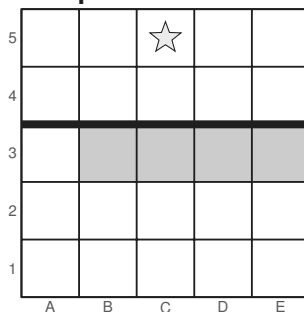
Shape 1



works: _____

jumps: _____

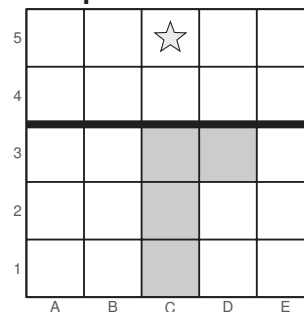
Shape 2



works: _____

jumps: _____

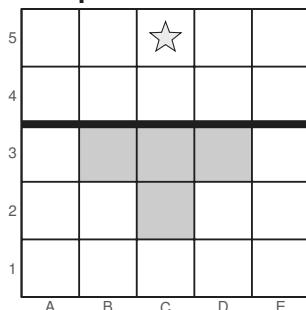
Shape 3



works: _____

jumps: _____

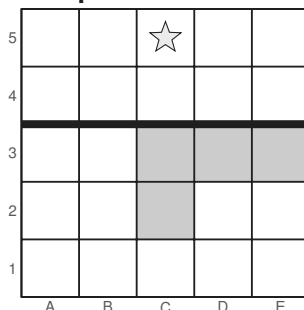
Shape 4



works: _____

jumps: _____

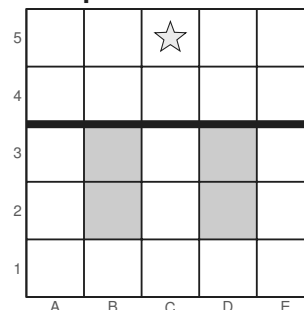
Shape 5



works: _____

jumps: _____

Shape 6



works: _____

jumps: _____

Problem 13: Board V shows a value in the corner of every square of board U. For each line, put pennies on the squares named, make the jump, and write the total before and the total after.

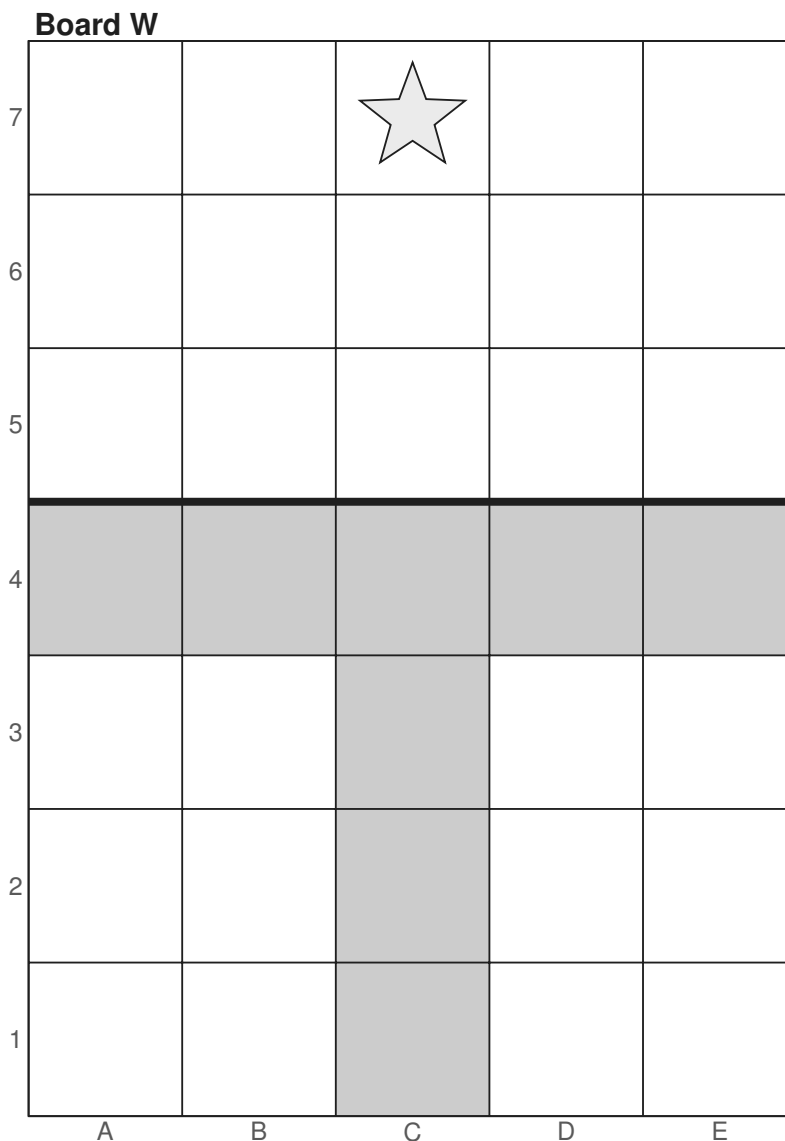
C3 and C2; jump up	before _____	after _____
D3 and E3; jump to the left	before _____	after _____
B3 and C3; jump to the right	before _____	after _____
C4 and C3; jump down	before _____	after _____

Board V

5	8	13	21	13	8
4	5	8	13	8	5
3	3	5	8	5	3
2	2	3	5	3	2
1	1	2	3	2	1
	A	B	C	D	E

Problem 14: On board V, find the three home squares with the largest values and write their total: _____. Write the value of the star: _____. Then write the total of the four squares of shape 1: _____ and of shape 5: _____. Tell the adult at your table what these totals say about Problem 11, and what shape 1 shows about whether a total of 21 is enough.

Problem 15: Board W has eight pegs: five along the river and three below the middle one. Put pennies on the gray squares. Find seven jumps that put a peg on the star at C7. Write down each jump as you make it: the square the peg jumps from, the square it jumps over, and the square it lands on. If you get stuck, set the pegs up again and write a fresh list.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

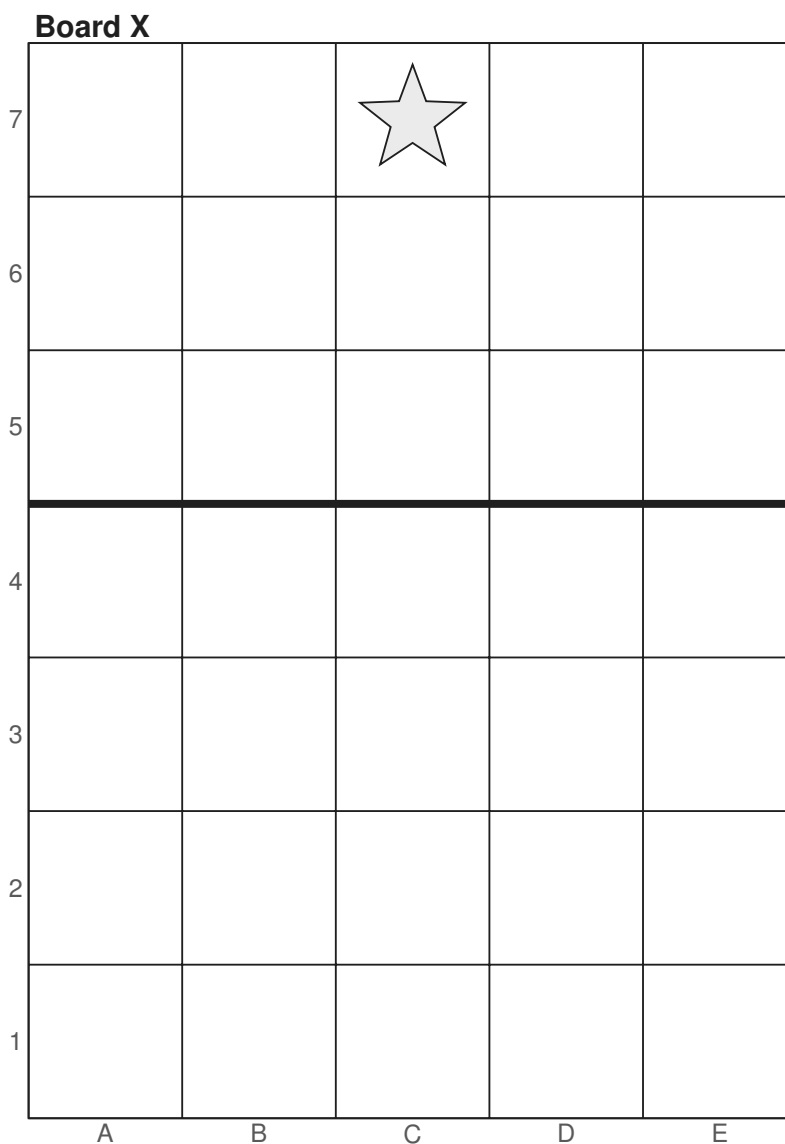
Jump 6: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

Jump 4: from _____ over _____ to _____

Problem 16: Board X is empty. Find a set of eight home squares, different from the shape of board W, from which a peg can reach the star at C7. Test it with pennies. Draw a dot in each of the eight squares of a set that works, and list its seven jumps.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

Jump 6: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

Jump 4: from _____ over _____ to _____